Switch-Controlled LED Sequencer (Ascending & Descending)

Project Overview

This project demonstrates a bidirectional LED chaser system controlled using two switches. One switch activates an ascending LED sequence, while the other activates a descending sequence. When neither switch is active, all LEDs turn off.

Objective

To design a dual-switch-controlled LED chaser system capable of running LEDs in ascending and descending sequences.

Hardware Requirements

1. Arduino board
2. 8 LEDs
3. 2 Push switches
4. 220Ω resistors
5. Breadboard and jumper wires

Pin Configuration

LED Pins: 13, 12, 11, 10, 9, 8, 7, 6

Switch Pins: A0 (Ascending), A1 (Descending)

Working Principle

Pressing Switch 1 runs LEDs from LED1 to LED8. Pressing Switch 2 runs LEDs from LED8 to LED1. If no switch is pressed, all LEDs remain off. Each LED transition uses a 500 ms delay.

Applications

1. Embedded systems learning
2. LED animation demonstrations
3. Microcontroller lab exercises
4. Educational training kits

Future Enhancements

1. Variable speed control
2. PWM fading effects
3. Multiple lighting patterns
4. Interrupt-based switching